



A model for improving the role of IT in BPR

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Abstract

Purpose – Extensive literature on business process management suggests that organisations could enhance their overall performance by adopting a process view of business. However, there is a lack of empirical research in this field. The purpose of this paper is to investigate the understanding of the process view and process maturity levels in a transition economy and to test the impact of process orientation maturity level on organisational performance.

Design/methodology/approach – Empirical investigation combined an exploratory-confirmatory approach using factor analysis and structural equation modelling.

Findings – The investigation confirms the impact of business process orientation on organisational performance in a transition economy. The link is even stronger than in the original investigation. The results show that business process orientation leads to better non-financial performance and indirectly to better financial performance.

Practical implications – The research confirms that business process orientation is advantageous for companies since it has a positive influence on organisational performance. The finding that the impact on financial performance is indirect through non-financial performance suggests that the companies have to take that view of performance into consideration as well.

Originality/value – The paper is valuable for academics and practitioners because the impact of business process orientation on organisational performance has been confirmed for a transitional economy. Its originality is in the measurement of organisational performance, for which a more detailed specification of organisational performance based on the balanced scorecard concept that includes non-financial performance measures has been used.

Keywords Process management, Organizational performance, Financial performance, Business process re-engineering, Communication technologies

Paper type Research paper

1. Introduction

The original approach to business process re-engineering (BPR) (Hammer and Champy, 1993) has caused problems for companies that attempted to undertake programmes to implement such “pure” BPR in real-world organisations (Silvestro and Westley, 2002). These problems may have been caused by those organisations having failed to view BPR as a part of a greater transformation process or by the planners failing to take account of the present role of information technology (IT) within their organisations (Attaran, 2003). Hammer (1990) originally advocated a “clean sheet” business-driven approach to BPR and this may have led to an underestimation of the role that IT plays in BPR. Even those authors who advocated a systematic analysis of IT provision before engaging in BPR may not have engaged sufficiently with the phenomenon (Willcocks, 2002). For instance, they do not always categorise the different roles that IT may have in BPR, nor do they investigate in detail the impact

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that these roles may have on the success of a BPR project (Willcocks and Smith, 1994). Although the amount of BPR research has reduced since the middle of the nineteen-nineties, the topic holds continued interest (Attaran and Wood, 1999; Al-Mashari and Zairi, 2000a,b) and it can be argued that is now more important than ever to understand the role of IT in BPR (Wu, 2003) as the relationship between IT and business processes and structures is becoming increasingly complex. In addition, academic and industry interest in the adjacent area of strategic alignment has been growing for more than a decade (Coleman and Papp, 2006). The developments proposed in this paper build on but differ from existing strategic alignment frameworks (Henderson and Venkatraman, 1993) and are an addition to existing BPR methods. This makes it worthwhile to revisit the topic of IT evaluation as a preparation for business change.

2. Methodology and structure of the paper

Clearly, the relationship between IT and BPR is complex, and a suitable definition of BPR must be produced before that relationship can be understood. A literature survey leads to a redefinition of BPR, and based on this redefinition the paper goes on to categorise, describe and discuss the possible roles of IT in BPR. Attention is also drawn to some IT issues that are important to any BPR project, particularly where connectivity and flexibility are involved. The paper also considers the issue of strategic alignment (Avison *et al.*, 2004) between the business and IT strategies and proposes an evaluation framework that has three purposes. First, it supports IT planners in identifying the likely role of an organisation's IT infrastructure in relation to BPR. Second, it provides business planners with guidance in assessing the organisation's capability to undertake a BPR programme. Third, both groups of planners can use the framework to examine the relationship between the organisation's present IT capabilities and its potential to carry out BPR projects. The project can then be planned and implemented in a way that takes into account the present and future roles of IT in BPR, which should improve the probability of success. The final part of the paper comprises a critique of the framework and concludes by offering suggestions for future research in this area.

3. The background to BPR

To understand the different views of BPR it is first necessary to define the word process (Davenport, 1995; Earl, 1994). A process has been defined as "a collection of activities that takes one or more kinds of input and creates an output that is of value to the customer" (Hammer and Champy, 1993, p. 35). Dhillon and Hackney (2003) call a process "a set of logically related tasks performed to achieve a defined business outcome". Davenport (1993, p. 5) refines this definition to "a specific ordering of work activities across time and place, with a beginning, an end, and clearly identified inputs and outputs; a structure for action". These definitions are very general, and any of them may apply to a typical business process such as order fulfilment in a typical commercial organisation. The most generalised view is that "processes are...how things get done" (Earl, 1994, p. 13). It is possible to identify several approaches to BPR, which approximate to "generations" and which serve to illustrate the changing views of BPR over time. The first approach views BPR as "radical restructuring", as shown in the following definitions:

The fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical, contemporary measures of performance, such as cost, quality, service and speed (Hammer and Champy, 1993, p. 32).

The fundamental rethinking and redesign of operating processes and organisational structure, focussed on the organisation's core competencies, to achieve dramatic improvements in organisational performance (Lowenthal, 1994, p. 62).

In this view re-engineering first determines what a company must do, then how to do it. It ignores what is and concentrates on what should be. In determining what a company should be doing, the actual business strategy or vision of a company may also be subject to scrutiny, and possibly to change. However, the emphasis is on radical process redesign, which challenges the bureaucratic assumptions behind the processes and invents new ways of working that are completely different from those of previous eras (Kawalek, 1994; Towill, 2001). Such radical re-engineering should be undertaken when a "quantum leap" in business process improvement is required, and a breakthrough in performance cannot be achieved by simply redesigning existing processes for this is "merely rearranging the deck chairs on the Titanic." (Hammer, 1990, p. 107). A "clean sheet" approach to BPR is being advocated here, in which existing business processes and structures are set aside or purposely disregarded. This approach is intended to ensure that the new process is not "tainted" by old habits and assumptions, the so-called "paving over cowpaths" syndrome (Hammer, 1990), but some authors advocate analysing the old processes as the key first step in a BPR project (Gunasekaran and Kobu, 2002).

Through this analysis the BPR team gains a deeper and more fundamental understanding of the underlying objectives of the system that the processes are intended to support. This may be especially important if external consultants (who may lack "local knowledge") are driving the BPR project, since these objectives may be implicit or "cryptic" rather than stated or open. Davenport (1993) makes a case for examining an organisation's current IT infrastructure to recognise the constraints it may impose on the redesign of the business processes. In this view appropriate metrics (e.g. activity based costing or cycle time analysis) should be applied before and after a BPR implementation, to allow the company to judge the success of its BPR projects (Kuwaiti and Kay, 2000). It is also preferable for a company to set objective performance levels by benchmarking (Johnston *et al.*, 2001). In this "less radical" view of BPR the existing processes and structures are regarded as valuable benchmarks and as indicators of the organisation's purpose and culture.

3.1 Different views of BPR

The emphasis in the less radical view of BPR is on the systematic aspects of how things are done as in other approaches such as total quality management (TQM) and process improvement (Davenport, 1993; Earl *et al.*, 1995). In these approaches, the aim is to streamline the process in the organisation's value chain in order to add value incrementally (Johansson *et al.*, 1993). In BPR the goal is to replace the whole process with one that is "much more effective for both the customer and the organisation itself" (Robson and Ullah, 1996, p. 6). This is intended to shorten lead times and reduce bureaucracy, which in a typical application (e.g. an order fulfilment process) should mean that requirements will be met more quickly and with less frustration. This should in turn result in lower costs, better service levels and consequently the

opportunity to improve competitiveness and increase market share. These effects could also be achieved through TQM or process improvement as well as BPR, but with BPR the targeted improvements will be more significant. This approach to BPR is essentially similar to certain aspects of the “systems approach” (Couger *et al.*, 1994) as the interdependence of tasks, roles, people, departments and functions is highlighted and cross-functional thinking is encouraged (Davenport, 1993; Earl, 1994). It is possible to take this “systems view” of processes one step further, as a business process such as order fulfilment “requires connectivity between the traditional application systems of purchasing, production, inventory control, distribution and sales order processing” (Moreton and Chester, 1997, p. 144). The integrated IT infrastructures of many modern organisations are ideally suited to provide such connectivity, both within organisations and between organisations through “webs of value” (Selz, 1998; Al-Mashari and Zairi, 2000a,b).

Despite not being emphasised in many of the common definitions of BPR, IT clearly has an important place in BPR projects (Khalil, 1997; Bhatt, 2000). Although their original (i.e. “radical”) definition of BPR does not explicitly mention the role of IT, Hammer and Champy (1993) point out later that IT is not only “crucial” to BPR but, when used creatively, IT enables process redesign to be increasingly radical in scope and scale. This may bring the “radical” and “conservative” approaches closer together by making radical BPR projects more feasible technically. Conversely, if misused, “technology can block re-engineering altogether by reinforcing old ways of thinking and old behaviour patterns” (Hammer and Champy, 1993, p. 83), while at the same time giving them the appearance of modernity. Clark and Stoddard (1996) conclude that BPR is a combination of process and technological innovation and that a merger of technological and process change is needed if major benefits are to be gained. A summary of the range of definitions of BPR and its characteristics is given by Choi and Chan (1997, p. 40).

4. Problems with BPR in practice

A major problem with BPR when the radical or “clean sheet” approach has been adopted is that not many organisations can afford to “obliterate” their present infrastructure and to implement a completely new one, nor can they afford to interrupt their business while its core processes are re-engineered (Attaran and Wood, 1999). In practice, a general rule for re-engineering is therefore “revolutionary design, evolutionary implementation” (Lockamy and Smith, 1997; Bustard and He, 1998). The case study of Capital Holding in Hammer and Champy (1993) illustrates the use of an evolutionary approach through the introduction of pilot schemes in what is referred to as Company B, the new company. This co-existed with the old company (“Company A”) during the implementation period, which it was envisaged would take five years in total. Yet another difficulty facing companies wishing to undertake a BPR programme is knowing which processes to redesign. The existing BPR approaches give little practical advice on process selection, and criteria for prioritisation are rarely offered in the BPR literature, although it is often argued that better results are gained by the selective approach (Berztiss, 1996; Johansson *et al.*, 1993; Robson and Ullah, 1996). Hammer and Champy (1993) suggest three criteria for process selection. First, dysfunction, which means prioritising those processes that are the most troublesome, second, importance, which suggests concentrating on the processes with the greatest

impact on the company's operations and, finally, feasibility, which identifies those processes that are the most susceptible to successful redesign.

4.1 BPR and competitive advantage

Other criteria are suggested by Robson and Ullah (1996), who suggest selecting those processes where competitors are forging ahead, or where real competitive advantage might be gained through the application of BPR. Most importantly, Johansson *et al.* suggest concentrating on "those processes that the business's strategy has identified as critical to... match or beat the competition" (Johansson *et al.*, 1993, p. 59). This is similar to key performance factor or key result area analysis (Peffer *et al.*, 2003) and makes explicit the link between BPR and business strategy explicit. This link is essential, as BPR should ideally be strategy-led (Guha *et al.*, 1997) and strategic planning has an important role in BPR programmes (Berztiss, 1996). It seems, therefore, that the initial definition of BPR as given by Hammer and Champy (1993) is rather narrow. The main failing of this definition is that it does not explicitly recognise BPR as a part of an organisation-wide transformation. BPR cannot deliver "dramatic and sustainable performance improvements" unless it is augmented by changes in its culture, structure and technology, including the IT infrastructure and applications (Kettinger and Teng, 1998). Studies in the USA suggest that globalisation, time to market, risk management, service and cost would tend to push organisations into mutually dependant relationships relying on effective IT support (Bhatt, 2000). As these forces became more compelling through the growth of electronic data interchange (EDI) and electronic commerce, organisations linked up more closely for their mutual interest. As a result, processes now routinely extend beyond the organisational boundary to include the external value chain (Normann and Ramirez, 1993). For example, the order fulfilment process may link a company to its suppliers and customers through a shared IT infrastructure.

4.2 BPR and flexibility

This leads back to the issue of connectivity, but on a wider scale and in a different context. IT has become an important connecting factor and has been used to integrate internal functions and departments into the business processes of an organisation, as well as external customers and suppliers (Kim and Narasimhan, 2002). The role of IT in BPR is therefore as significant as that of strategy, and it follows that alignment must exist between the re-engineering initiative, the business strategy and the IT strategy (Lockamy and Smith, 1997). This suggests that changes in the IT infrastructure which will almost certainly have an impact on business strategy, as the two must be kept aligned to obtain maximum organisational effectiveness and efficiency. This leads to the conclusion that IT will inevitably be of strategic value to a company that is involved in business change (Avison *et al.*, 1996) and that a flexible IT infrastructure is needed to facilitate change when the need arises (Avison *et al.*, 1994; Fitzgerald and Siddiqui, 2002). In an "ideal" organisation, business processes would be reassessed and re-aligned continuously with a view to improving, changing or removing them. This would reduce the need for wholesale re-engineering every few years by ensuring that the organisation's infrastructures and strategies are in alignment (Grant, 2003). This is described as an "evolutionary" (rather than a "radical") approach to BPR (Al-Mashari and Zairi, 2000a,b).

5. BPR – a refined definition

As the preceding discussion illustrates, the original definition of BPR may be improved. The following redefinition of BPR is therefore suggested:

BPR is the fundamental rethinking and radical redesign of appropriate business processes to achieve dramatic improvements in critical, contemporary measures of performance, such as cost, quality, service and speed. Such redesign and pace of implementation to be suited to the individual organisation, contingent upon the “gap” between the present state of the organisation’s structure, culture and IT infrastructure, and the state required to implement the new business processes successfully. An ideal state would be one in which BPR was an ongoing, proactive process.

“Appropriate” in this definition applies that not all business processes will yield the dramatic improvements sought by an organisation and that each organisation will need to find those particular processes that lend the greatest support to its business. This alignment of BPR with the business strategy is essential to the success of any BPR initiative (Al-Mashari and Zairi, 2000a). The scope and scale (and probably the speed) of the redesign project will depend on the extent of the gap between “what is desired” and “what is current”. To achieve “what is desired”, the organisation should use an “evolutionary” type of BPR (Johnston *et al.*, 2001), supported by an appropriately flexible IT infrastructure (Avison *et al.*, 1997; Fitzgerald and Siddiqui, 2002) to allow the changes to take place. This implies that the role of IT in the organisation may be vital to the success of that organisation’s BPR strategy and to exploit this fact it is important to understand the role that IT plays within the organisation’s BPR effort. This is the subject of the next section of the paper.

6. The roles of IT in BPR

Based on the revised definition of BPR, it is necessary to determine the role that IT may play in implementing BPR in a typical organisation. Hammer and Champy (1993) identified a number of IT applications or technologies that they called “disruptive technologies” that were supposed to challenge traditional business processes and structures and stimulate business change. In the last decade a number of these technologies have developed in a way that confirms their relevance to the current research study. The original list of “disruptive technologies” includes the following:

- Shared databases and integrated systems (e.g. enterprise resource planning (ERP)), which make information available in many places from a single source and therefore need to follow “end-to-end” processes (Presley, 2006).
- Knowledge-based systems, which allow generalist “knowledge workers” to perform specialist tasks and decision-making to be delegated in support of organisational objectives (Hendriks, 1999).
- Telecommunication and wireless communication networks that allow organisation structures to be decentralized and enable work to be distributed (Wei *et al.*, 2006).

In addition, we may now include web-based systems that may be combined with any of the above technologies and logistical information systems such as RFID and automatic identification and tracking systems that often have an integrating function not just within organisations but between organisations (Bergeron *et al.*, 2005).

In order to determine the possible roles of IT, a review of the relevant literature (including case studies) was undertaken using the revised definition of BPR as given in the previous section as a basis. Existing work in the area (Chan, 2000; Gunasekaran and Nath, 1997; Al-Mashari and Zairi, 2000b) was also examined to see if parallels could be established. It was found that IT could have the role of a constraint, a driver, could be neutral or a catalyst, or could be an enabler or be proactive. The six different roles of IT in BPR are defined in detail in the following section and their characteristics are shown in Table I.

6.1 IT as a constraint

The identified role of IT as a constraint is defined as “those aspects of existing technology infrastructure that limit the possibilities for innovation and cannot, for whatever reason, be changed in the relevant time frame” (Davenport, 1993, p. 50). In other words, certain kinds of IT can interfere with or prevent BPR from taking place. This often refers to the existence of legacy systems, where it would be too expensive or too disruptive to replace the IT infrastructure to support the redesigned business processes (Avison *et al.*, 1997). Maintaining existing business processes in this way will ultimately limit the ability of an organisation to pursue its strategic vision (Avison *et al.*, 1998) and can lead to various kinds of problem at an operational level, such as “islands of automation” caused by lack of IT compatibility and duplication. Davenport (1993) gives as an example a company with 37 different billing systems. Each organisational unit had created its own system in order to understand its individual contribution to the company’s revenues and profitability. This company was having

Role of IT	Characteristics of the role
Constraint	Legacy IT systems dominate main business processes. Inflexible IT infrastructures. Lack of skill and/or investment in new IT. Business processes embedded in existing IT systems. Lack of potential for investment in IT due to budgetary factors. Lack of perception of the potential of IT by management. Strategic alignment is low
Catalyst	New IT has been acquired. Changes in the business have been made that favour the use of IT. New management that sees the potential of IT in business change. New relationship developed with IT vendor, consultant, or service provider. Strategic alignment at crucial stage
Neutral	Lack of IS applications and IT infrastructure in the organisation. No IS or IT strategy in place. Business change targets are not well defined. The business is in an industry with low information intensity or little competition through IT. Strategies and infrastructures are in alignment
Driver	The business has technological capability and seeks to exploit it through business opportunities. Possibly a new business or a technological innovation. Sufficient investment is available and IT development is not a limiting factor. Strategic alignment process is proceeding rapidly
Enabler	IT is a key performance factor and a “competitive arena” in the industry. Management has a clear business vision and a future change plan. Business change targets are well defined. Sufficient investment is available and IT development not a limiting factor. Strategic alignment in process
Proactive	Management has a clear business vision and future change plan. The IS and IT infrastructure are well developed. There are few constraints on IT development. Management sees the potential of IT. Strategies and infrastructure are in alignment

Table I.
Characteristics of the role
of IT in BPR

difficulty just attempting to streamline its order management process due to the constraints imposed by its incompatible IT platforms.

Another example of such an occurrence is the Inland revenue (IR) in the UK. Prior to engaging in a BPR effort, IR had made a number of major investments in IT, but these had resulted in the existence of a variety of platforms and the usage of large systems, each having different standards, conventions and protocols:

The existing systems are very efficient at providing computerised procedures. However, they are also inflexible, slow and expensive to change and are all relatively self-contained, resulting in large-scale duplication of data (CCTA 1994, p. 3).

IR therefore developed a policy of “introducing change through islands and pilots” (CCTA, 1994, p. 8). This is another example of “evolutionary” BPR, where we see an organisation with a constraining IT infrastructure being forced to invest gradually in a new “enabling” IT infrastructure in order to redesign its business processes. Even with integrated systems there can be a problem with flexibility, as the linked processes may be too inflexible to support radically process change (Avison *et al.*, 1997). Davenport (1993) gives the example of Bell Corporation in the USA. When strategic planners at a regional Bell operating company were considering a radical change in key business processes, they assumed that all their systems would be purchased “in house” (from its Research and Development wing) rather than by adopting “industry standard” IT packages. The planners were unaware of the conflict between their goal and the compatibility constraints imposed by the systems adopted by its customers and suppliers. The whole BPR programme at Bell was therefore compromised by its “imposed” IT strategy (Mitchell and Zmud, 1995). In summary, the role of IT as a constraint can be identified where IT prevents, limits or inhibits the successful implementation of BPR programmes. IT in this role is referred to as an inhibitor (Zairi and Sinclair, 1995).

6.2 *IT as a catalyst*

Another identified role for IT can be defined as that of catalyst or “initiator” of BPR (Chan, 2000). In this role the technology can trigger the business change, for instance an IT project that affects business processes (e.g. updating an office IT platform) could be seen as an opportunity to redesign those processes (Dibrell and Miller, 2002). Chan (2000, p. 227) discusses work relating to the case of computer imaging technology changing business process processes in a hospital. A second possibility is that the technology is already in place but that the processes are not yet able to take full advantage of its available capabilities, for instance due to lack of IT knowledge on the part of the process owners. In such cases the IT department needs to understand the business processes and see how they could be improved, and needs to communicate this vision to the management of the organisation. In this way “a business process that maximally exploits the available IT capabilities is developed, and we begin to see an evolving alignment between technology and organisation” (Venkatraman, 1991, p. 137). The third possibility is more sweeping, in that newly developed technology sometimes “makes feasible activities and actions of which people have not yet dreamed” (Hammer and Champy, 1993, p. 88). In such cases IT may become a “strategic catalyst” (Stoddard and Jarvenpaa, 1995, p. 82), although “the challenge that most corporations fail to meet is recognising the business possibilities that lie latent in technology”

(Hammer and Champy, 1993, p. 88). Individuals can also have a key part in reinforcing the role of IT as a catalyst, and Earl observes that “bright spark commercial managers backed by product champions” had exploited IT effectively in creating business change in some very successful organisations (Earl, 1989, p. 77). In summary, this role is identified where IT triggers business change directly or indirectly (O'Donoghue *et al.*, 2003) but is not directly used in the change process.

6.3 The neutral role of IT

This has also been referred to as the “facilitator” role of IT (Davenport, 1993, p. 200; Chan, 2000, p. 227). The definition used here is that IT is “neutral” in relation to the BPR project, as IT does not in itself directly affect an organisation's ability to re-engineer its business processes successfully, but acts as a simple tool for supporting the BPR effort. Kettinger *et al.* (1997) suggest that in this role IT may be used throughout the BPR lifecycle to support various activities such as:

- *Defining the business strategy and process vision* – IT can be used for the collection, analysis and dissemination of information from various sources and also allows groups to collaborate efficiently and effectively (Avison *et al.*, 1998).
- *Understanding the current process* – IT-based process flowcharting tools are useful for arriving at and documenting a common understanding of the structure of the current process(es).
- *Analysing and evaluating both the current and the new process* – IT may be used to calculate activity-based costs, assist in conducting surveys or run simulations to ascertain potential bottlenecks. Software re-engineering may be used to determine the underlying logic and functionality of an old system which may then form the basis for a new system (assuming that it supports the new process in a satisfactory manner).
- *Designing the new process* – IT-based process modelling tools facilitate the creation of informative process flow charts, and may also be used to design and run a model of the new process so that the two may be easily compared. A sophisticated tool may support sufficient levels of detail to be useful during the systems design/prototyping stage.
- *Prototyping the new process* – the use of fourth generation languages, object oriented languages with class libraries, programmable database packages and code generating CASE tools has meant that prototyping can now be carried out more rapidly., meaning that the user requirements as captured will be more current (Avison *et al.*, 1997).

The neutral role of IT can be identified in cases where IT is used as a tool in the BPR programme, or when the resulting processes use IT in the “support” or “key operational” roles (Earl, 1989). It differs from the catalyst role in that it may be used as a support for the new processes (i.e. the results of change), but does not trigger the change itself.

6.4 IT as a “driver”

In this instance the term “driver” is used to typify the phenomenon of “technology push” (Bughin, 2003). In some instances new technologies are introduced rapidly into

a company, in the mistaken belief that being the first to implement such technology will keep them ahead of the competition (Dhillon and Hackney, 2003). In some ways, the BPR project is undertaken in an attempt to exploit the potential of a new technology (perhaps as a response to competitive pressure). In such cases, there may be no understanding of the relevance, if any, to the organisation's overall strategy or business objectives. In such organisations, processes may be re-engineered to fit in with the new technology (Marchand and Stanford, 1995; Davenport, 1993) such as the introduction of ERP systems (Ramaprasad, 1999). There are two possible problems with this. First, the redesigned process may be less than optimal in terms of process efficiency. Second, strategic misalignment may occur, as the new process may not contribute to the achievement of strategic business objectives (Avison *et al.*, 1998). This in turn could lead to a full or partial "failure" of the BPR effort. Another example of IT as a driver may be observed if the BPR programme is initiated by the IT department.

This is illustrated by the example of a BPR programme initiated by the Paper Company. In this case BPR was seen primarily as an opportunity to introduce new technology to bring the company up to date and to solve a number of business problems that had been noticed, generally concerning difficulties with accessing information from diverse parts of the organisation (Guha *et al.*, 1997). Problems arose when the IT department failed to elicit support from the various organisational functions affected by the necessary reorganisation. The perception in the company was that the BPR project was being driven by technology and "had more to do with IT planning than with corporate-wide business planning" (Guha *et al.*, 1997, p. 132). This perception led in turn to a lack of support throughout the organisation in general and to a failure of the BPR programme. Another company, Comdisco, experienced similar problems, but in their case the BPR project was reoriented towards the business side of the company and eventually the initiative was given the necessary support and succeeded. The message here is that BPR programmes should not be driven by IT, but should be guided (and should be seen to be guided) primarily by an organisation's strategic thrust and direction (Guha *et al.*, 1997), otherwise the BPR programme is likely to fail. In summary, the role of IT as a driver can be identified where IT developments are the reason for the redesign of a business process. For example, a large-scale IT application, typically an ERP system or integrated manufacturing system (Saad, 2003) may be installed in an organisation and the existing business processes and structures may then need to be redesigned to ensure compatibility with the new application. It differs from the catalyst role of IT as in the "driver" role IT assume a greater purpose and importance in the BPR project.

6.5 IT as an enabler

In the literature IT is commonly described as "an enabler of BPR", but it is not always clear exactly what this means, or how it differs from the role of IT as a catalyst. Chan points out that this role has received the most attention in the literature (Chan, 2000, p. 228). Davenport (1993, p. 50) describes the role as "using technology in ways new to the company or industry to achieve process innovation". Alavi and Yoo (1995) cite the use of IT networks in enabling parallel data access, analysis and processing. Davenport (1993) divides the enabling role of IT in BPR into nine different categories ("automational", informational, sequential, tracking, analytical, geographical, integrative, intellectual and "disintermediating") based on the impact that the

technology would have on process innovation. Hammer and Champy (1993, p. 83) observe that IT “permits companies to re-engineer business processes” in radical new ways that “exploit the latest capabilities of technology to achieve entirely new goals” (Hammer and Champy, 1993, p. 85). One example of the enabling role of IT is the use of EDI by Her Majesty’s Stationery Organisation (HMSO) to improve the management of its supply chain for UK Government publications (CCTA, 1994, p. 3) as a part of its strategy for service improvement. The transaction handling had to be transformed, dramatic improvements in the error rate and response times were required, along with increased customer satisfaction and reduced costs per unit. HMSO recognised that EDI could play a key role in achieving these objectives, in conjunction with corresponding changes in the business processes. They not only developed an electronic catalogue and ordering system called HEART (HMSO enquiry, access and rapid trading) containing descriptions and pricing information of their products, but they also trained customers to place orders electronically and supplied them with a highly usable software package. HMSO orders grew from being worth £4 million in 1991 to a value of £45 million in 1993. This rapid growth reflected not only an increase in the number of orders, but also a growth in the number of customers using HEART. At the other end of the chain HMSO was trading electronically with 130 key suppliers. In addition to this, when an order was filled from stock it automatically led to a replacement order, and this in turn was passed through to the supplier via HMSO’s corporate electronic gateway. EDI had, in effect, become the primary means of order filling at HMSO (CCTA, 1994).

In the case of HMSO, IT enabled an increased scope and scale of operation. After the implementation of BPR, its EDI capability was freed from the limitation of a number of chosen partners (as is the case with many EDI applications) and had gained the potential to trade internationally, even though such a capability was not needed at that time. When the trading fund order was passed, allowing HMSO to trade with Europe, it was ready and waiting. From the very beginning HMSO was committed to an open network – its gateway can therefore “talk” to any possible customers and suppliers. In addition to this, it had the foresight to follow UN-sponsored standards in printing and stationary, instead of the limiting British standards. The enabling role of IT is continuous in this case. HMSO is actively searching for new technology that will benefit the business (e.g. the use of “electronic cash”) by supporting business processes that are yet to be developed. The “enabling” role of IT is identified where an organisation has clear plans for the redesign of business processes in support of its business strategy and seeks to develop its IT capabilities in line with its business plans through a process of strategic alignment (Luftman *et al.*, 1993). The main difference between the enabler role of IT and the driver role is that the former is a result of “business pull” and the latter of “technology push”. There are successful examples of the use of technology (e.g. ERP) in this role (Huq and Martin, 2006).

6.6 Proactive IT

This may be seen as the “ideal” role of IT in BPR. It is most likely to be found in an “organic” company – one that is at home with change and can deal with both the “complexity and volatility” of the business environment (Berztiss, 1996, p. 17). Such a company is likely to regard BPR as “ongoing”, and it will be “institutionalised” alongside IT (Caron *et al.*, 1994). From the case studies, it can be seen that

an organisation using IT proactively will behave in a predictable fashion. It will also seek to maximise its ability to improve, redesign or abandon relevant business processes and to implement the necessary changes through the use of IT. The “proactive” organisation will monitor its environment, actively looking for new opportunities or threats in terms of business and IT. Finally, it will seek to select the best option for dealing with a threat or an opportunity, and will use IT to counter or to exploit this threat or opportunity within the relevant time frame. The “proactive” role of IT is therefore multidimensional. IT will not only to support the redesigned business process, but will also influence the process of redesign itself, including the creation of new design options (Clark and Stoddard, 1996).

As Davenport and Short (1990, p. 12) observe, “in leading edge practice, information technology and BPR have a recursive relationship. Each is the key to thinking about the other”. The potential of IT in this role is not only to support business process redesign but also to make possible the higher levels of transformation (such as “business network redesign” and “business scope redefinition”) to take place should the opportunity arise (Venkatraman, 1991, p. 128). In undertaking BPR it makes sense to consider the IT issues that are relevant to these higher levels of transformation, even if they cannot yet be attained, as these are where the greatest benefits may lie (Caron *et al.*, 1994; Grover *et al.*, 1994). In the proactive role, IT should also have an “interdependent infrastructure” (Venkatraman, 1991, p. 151), whereby any changes in the IT infrastructure signal the possibility of changes and improvements in strategies, and *vice versa*. To enable the organisation to partake fully in the proactive role, the IT infrastructure and the strategic context should therefore be in a constant state of strategic alignment (Luftman *et al.*, 1993). This leads on to the necessity for a flexible IT infrastructure (Avison *et al.*, 1997) which “would be one that was designed to evolve, itself, with emerging technologies and would support the continuous redesign of business and related IS processes” (Duncan, 1995, p. 44). The importance of such flexibility to business is recognised in a variety of sectors, with examples such as the IR (CCTA, 1994), Sweden Post (Moreton and Chester, 1997, p. 147) and BP (Cross *et al.*, 1997). The proactive use of IT in BPR is often associated with the strategic information systems (Avison *et al.*, 1996) and the informed use of strategic alignment (Luftman *et al.*, 1993).

6.7 Discussion of the roles of IT in BPR

The roles of IT as described above may be visualised as being on a continuum, with the constraining role at the “negative” extreme and the proactive role at the “positive” extreme, as shown in Figure 1. This view reflects the degree to which the IT

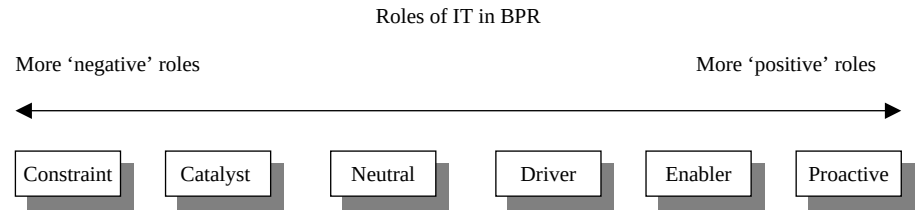


Figure 1.
A continuum of the
possible roles of IT in BPR

infrastructure (as it exists at present) is likely to constrain or to lead the BPR effort in an organisation. The role of IT in BPR therefore depends on two factors:

- (1) The type of IT infrastructure that the company has (e.g. whether “legacy systems” proliferate, or whether flexible IT platforms and open systems are in place).
- (2) The degree of strategic alignment (Avison *et al.*, 2004) that exists between the business objectives and the IT objectives (e.g. the extent to which organisational strategies and infrastructures are linked to the IT strategies and infrastructures).

It is also possible to conclude that the role of IT in BPR (through the IT infrastructure) depends on the IS strategy of the company and the degree of strategic alignment that the company is able to achieve. The usefulness of the IT infrastructure to the organisation reflects the degree to which it supports its business objectives at any given time. It therefore follows that when the business objectives of a company change, then the IT strategy must also change in order to support the new objectives. For example, in a situation of major strategic change, a company could find itself in a position where the functional strategies (e.g. marketing and production) and the corporate strategy in an organisation were aligned and would support BPR, but the IT infrastructure (i.e. the technology) had not yet caught up. The timing aspect is therefore crucial to the role of IT, as prior to the development of the functional and corporate strategies IT may have had a neutral role. It may also be relatively easy to give IT a neutral or facilitator role through upgrades and developments to the infrastructure. It is therefore in the organisation’s interest to identify, “cultivate” and influence the role that IT plays in strategic business processes (including BPR initiatives). When the place of IT in BPR is considered and the alignment of the business and IT strategies is viewed in the light of the above discussion, the following points emerge regarding the characteristics of the roles of IT in BPR:

- *Constraint* – the IT infrastructure lacks flexibility and “openness”. IT is not considered to be strategic, and the IT strategy and business strategy are not aligned (i.e. there is a “technology gap”). The IT infrastructure or applications will therefore tend to inhibit the development of new business processes or relationships.
- *Catalyst* – the IT strategy and business strategy are not presently aligned, but alignment is being sought. There may be a “technology gap”, but the organisation is seeking to exploit the potential of IT in its BPR programme through harnessing creativity, perhaps in an uncontrolled way. The chances of IT failing to support BPR are relatively high.
- *Neutral* – the relationship between BPR and IT is weak or unclear. The most identifiable role of IT is as a “tool” in the BPR effort, but the “business pull” and “technology push” factors are equally balanced and the IT and business strategies may be aligned.
- *Driver* – the IT strategy and business strategy are not aligned, the IT strategy is dominating the BPR effort. There may or may not be a “technology gap”, but the organisation is subject to “technology push”, perhaps through the influence of

vendors or consultants. IT infrastructure developments may be *ad hoc* or may produce serious business/IT misalignment.

- *Enabler* – the IT strategy is aligned with the business strategy, there is no “technology gap” and the IT infrastructure allows the business strategy to be enacted.
- *Proactive* – the IT strategy and business strategy are not only aligned but are interdependent. In the enabling role, the IT strategy supported the business strategy. In the proactive role, the IT strategy is a part of the business strategy.

The degree of alignment between the “business pull” factors (e.g. business strategy and organisational infrastructure) and the “technology push” factors (e.g. IT strategy and infrastructure) and the impact of this alignment on the role of IT in BPR is represented by the model shown in Figure 2.

This model builds on the idea of a continuum of IT roles and includes the established concepts of “business pull” and “technology push” as influences on the development of the IT infrastructure (Venkatraman, 1991). These factors represent the “technology direction” that the organisation adopts in developing its IT infrastructure, and are an important influence on the role of IT in BPR. The “business pull” approach implies that the organisation is proactive in seeking to exploit opportunities in its business environment, and that the consequent demand for IT support from the business functions leads to the acquisition of IT resources. The opposite “technology push” approach implies that the organisation scans the technology market place and acquires an IT infrastructure and applications that may (or may not be) exploited by the business units. The two approaches are not mutually exclusive as the IT strategy in some organisations will seek to achieve a constant balance of “pull” and “push” factors. In practice, most organisations will probably respond to both factors individually at different times. The fact that the “business pull” axis in Figure 2 has three columns may imply that it has greater importance to a typical company than “technology push”, as in the ideal form of organisation the business strategy should be supported by the technology, rather than the other way around.

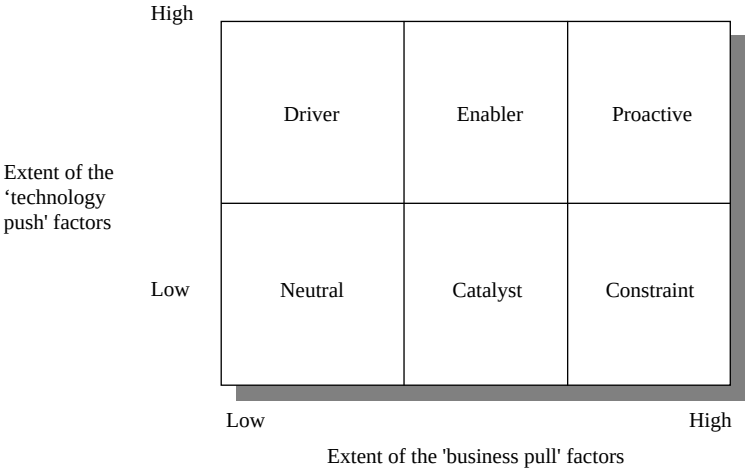
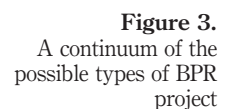


Figure 2.
Framework for linking
technology direction and
the role of IT

Based on the ideas put forward earlier, it is now possible to propose a useable framework for evaluating the role of IT in BPR. The objective of this framework is to identify in advance how the role of IT may affect the success or failure of the BPR effort, and it is suggested that the framework would be used by practitioners as a tool for analysis before and during BPR programmes. As such, it represents a useful contribution to the body of knowledge about the relationship between IT and BPR. The framework will serve to draw attention to the need for an organisation to ascertain the present role of IT in its operations, which includes investigating the present infrastructure and capabilities, and the degree of strategic alignment (Avison *et al.*, 2004) between the present IT infrastructure and business strategy. In addition, the BPR team will be able to tell from the framework what type of BPR they could hope to achieve under the present circumstances. For instance, if a company were currently committed to investing in “enabling” IT but had a “short-term” approach to acquiring its IT infrastructure, the framework would serve to draw attention to the longer-term outlook (i.e. the necessity to plan for a flexible infrastructure). The framework will also have use as an academic device for studying and explaining the relationship between IT and BPR. Other frameworks for explaining the role of IT or IS in BPR have been successful in this context (Chang and Powell, 1998). The literature search identified a number of “types” of BPR project in the organisations that were examined, which may be placed on a continuum from “failure” to “radical” in terms of their effectiveness in achieving the objectives of major business change (Figure 3). The ideas that form the basis of the proposed framework may be summarised by the following salient points:

- ### Types of BPR project



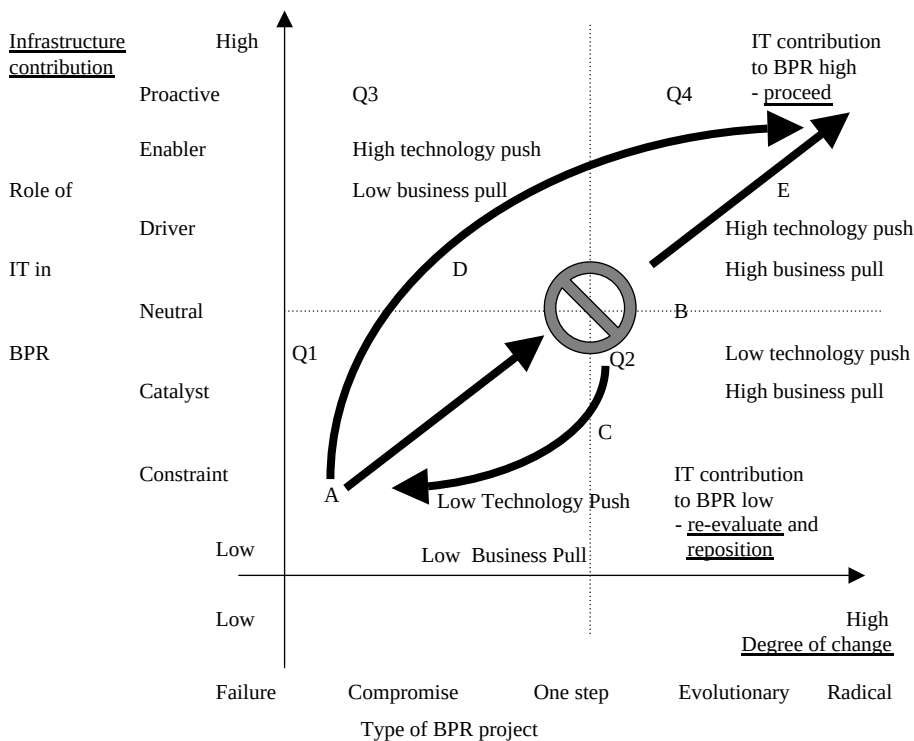
- The attainment of such a high degree of flexibility requires that the infrastructure should be very “open” or flexible. This gives a greater choice of suppliers, access to more customers, and more information on its environment. If data or applications are not shared it should be as a consequence of a business decision (i.e. emphasise the “business-pull” direction), rather than the organisation’s technological capability (Duncan, 1995).
- As “change” appears to be endemic in the business environment of today, the most effective type of re-engineered process would be one that was able to evolve and adapt to changes in requirements. This is the reason for identifying “proactive” IT as the most desirable role and for assuming that the more flexible the IT infrastructure is, the better it will support future BPR exercises.

There is a direct similarity between the concept of “range” and the type of BPR that is achievable, so the type is incorporated into the proposed model on the horizontal axis (Figure 3). A brief summary of the characteristics of each type of BPR project is given below:

- *Failure* – IT does not have a planned role in the BPR project, or the project has to be abandoned or is completed but fails to provide the expected business improvements.
- *Compromise* – In this type of BPR project the existing IT infrastructure cannot be changed within the given time scale. The BPR project has to take this into account and, although it will be a success, will be limited or unambitious in terms of its reach and range. Such BPR projects typically produce “islands of automation”, as they are applied to limited business processes or functions.
- *One-step* – The reach and range (and therefore the scope and scale) of a “one-step” BPR project are greater than for the former type, but the lack of IT support limits the potential of the newly designed processes for achieving the “higher level” transformations. In this case IT is not seen by an organisation as being truly strategic.
- *Evolutionary* – This is not the most radical type of BPR project, but the rate of business change will be incremental through targeted process redesign. The infrastructure will be sufficiently flexible to cope with the progressive change, and the IT strategy will be capable of maintaining alignment with the business strategy over time, although periodic adjustments will be required.
- *Radical* – This type of BPR project achieves major business change, with a high degree of reach and range, within an acceptably short time scale. The IT infrastructure is very flexible and copes well with the major “step change”, and the IT and business strategies are completely interdependent, being continuously in alignment.

The proposed framework represents the roles of IT in BPR and the type of BPR project and includes the concept of “business pull” and “technology push” (Figure 4) by superimposing upon the diagram four quadrants (Q1-Q4), which are interpreted as follows:

- Q1. *Low “technology push” and low “business pull”* – the “lower” roles of IT when applied to the “lower” types of BPR (i.e. to the left of the continuum) will be



A model
for improving
the role of IT

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Figure 4.
Proposed framework for
evaluating the role of IT in
BPR projects

associated with a generally low profile of IT in the organisation. Similarly, the commitment to radical change within the organisation may be poor. These factors will interfere with the organisation's ability to implement BPR successfully. For instance, IT is likely to be a "constraint" in an organisation that is aiming to achieve at best a "compromise" approach to organisational change in a forthcoming BPR programme.

- *Q2. Low "technology push" and high "business pull"* – the "lower" roles of IT when applied to the "higher" types of BPR will be associated with an organisation that has poor strategic alignment (Avison *et al.*, 2004). It may possess corporate ambition in planning a radical BPR programme, but its IT strategy will be weak (indeed, the whole IT function may be under-represented organisationally) and its IT infrastructure will lack flexibility and "openness". IT will therefore have a constraining role in an organisation that occupies this quadrant (e.g. typical symptoms will include "legacy systems" and "islands of automation") and will prevent the effective implementation of programmes of business change. This will occur irrespective of the organisation's ambition or competence in carrying out evolutionary or radical BPR.
- *Q3. High "technology push" and low "business pull"* – the organisation that occupies this quadrant will have a keen awareness of technological trends and standards, but a relatively poor business model. Its IT infrastructure will

probably be very advanced, and technology will have a high profile although the business model or strategy may be relatively weak or undefined. Such an organisation will exhibit poor strategic alignment (Avison *et al.*, 2004) (in common with the example in the previous quadrant), and may not be able to progress beyond a “compromise” or “one step” type of BPR, irrespective of its technical competence or resources. The potential of IT to be proactive or enabling in support of business change in this type of organisation will be wasted and the literature contains examples (Davenport, 1995) of high technology companies that failed to change as a result of poor business processes and plans.

- *Q4. High “technology push” and high “business pull”* – an organisation in this quadrant will combine a high profile for IT (e.g. well integrated “IT governance” and a flexible and open IT infrastructure) with a well-developed business plan and well-designed processes. It will be able to achieve a high degree of success in carrying out evolutionary or radical BPR, fully enabled by a proactive IT strategy. It is suggested, however, that success is not likely to come about by being competent in the isolated functions of IT and business. In order to occupy this quadrant, an organisation would need to achieve a high level of strategic alignment (Avison *et al.*, 2004).

7.1 Using the proposed framework

It is suggested that BPR planners could use the proposed framework in the following manner. First, there would be an “information audit” (probably undertaken by business analysts) to identify the probable type of BPR project that the organisation could support (Alexopoulos and Theodoulidis, 2003), which would be identified on a continuum of “failure” to “radical” using Figure 3 as an *aide memoir*. It is assumed that the organisation’s planners would not aim to produce a “failure”, but this may be a realistic appraisal of the likely outcome of a BPR programme where the organisation recognises its failings in advance. The additional factors that the organisation would need to assess in appraising its potential for achieving a particular type of BPR would include its ambition for change, its business analysis skills and experience, and the level of its resources. Also, at a higher level, the quality of its strategic planning processes and the degree to which its strategies and infrastructures are aligned would need to be taken into account. At the second stage of the process, the organisation’s IT planners would carry out an “IT audit” (Rau and Bye, 2003). This would entail a realistic evaluation of the likely role of the organisation’s IT infrastructure in the BPR programme, using Figure 3 as an *aide memoir* and Figure 4 as a framework to plot the organisation’s current position with regard to the forces classed as “technology push” and “business pull”. Other factors that would be taken into account in deciding a role for IT include the strength and position of the organisation’s IT management function and the flexibility or “openness” of its IT infrastructure. This would identify the probable role for IT in the type of BPR project that is being planned.

Taking into account both sets of factors, as a third stage in the process the planners can produce a plot on the proposed evaluation framework (Figure 4). This will allow them to examine the relationship between the organisation’s capacity for business change and its IT capability. Understanding this relationship is important in maximising the potential of IT in the planned BPR project. As an example, the organisation may assess its capability to achieve a given BPR project as being of the

“compromise” variety, with IT likely to act as a constraint (i.e. in *Q1*). Minor efforts to improve the BPR process and the IT infrastructure may allow the organisation to follow the path indicated by Arrow A, but it will ultimately be limited by its inability to progress beyond the “no entry” point of “neutral” IT in a “one step” BPR programme (Point B). This will be the case no matter how often BPR is attempted, as project will be forced to follow the path shown by Arrow C, meaning that the organisation must “return to the beginning” and repeat the process again. It should be noted that if the role of IT is “constraining” at this point and no change is effected, the IT infrastructure will continue to be a constraint the next time that BPR is attempted.

Clearly, the framework suggests that the organisation needs to bypass the “no entry” point by enhancing the role and potential of IT (i.e. Arrow D) from “catalyst” to one of the high “technology push” roles (i.e. those in *Q3*). This can be done by taking an entrepreneurial approach to IT management, by carrying out improvements to the flexibility and openness of the IT infrastructure and by ensuring that a high degree of strategic alignment (Avison *et al.*, 2004) is achieved. When the IT infrastructure is capable of supporting one of the “driver”, “enabler” or “proactive” roles, the organisation may be able to achieve full value from IT when implementing BPR (i.e. following Arrow E) by making its approach to BPR more radical.

8. Implications of the work

The proposed framework is needed because some organisations attempt to undertake BPR or strategic alignment without giving due consideration to the role of IT in BPR. Indeed, it has been observed that the “exclusion and expulsion of IS specialists from BPR programmes is a recurrent theme” (Markus and Robey, 1995, p. 592). The framework demonstrates that no serious BPR effort can afford to ignore the role of IT, and if it does so then the risk of failure is high. It appears then, that the role of IT in BPR is undervalued at present, especially in terms of its wider and more long-term implications. These implications are as follows. First, the IT strategy and business strategy need to be aligned for maximum benefits to be realised. Second, the IT strategy dictates the type of IT infrastructure within a company and third, the strategy and infrastructure should both support the business strategy and even influence it. Most importantly, as change is endemic to corporate life, then the IT infrastructure needs to be flexible in order to cope with changes in the environment and the business strategy. A flexible IT infrastructure therefore appears to be an increasingly desirable objective for companies in a rapidly changing environment (Avison *et al.*, 1997). By enabling an organisation to exploit potential business opportunities quickly, such flexibility helps give it competitive advantage (Duncan, 1995) and allows a quicker response to competitive threats. A flexible infrastructure should therefore be a future goal, especially if BPR is to be evolutionary (and therefore less disruptive), and this issue should be addressed specifically when planning to invest in a new IT infrastructure in support of a planned BPR initiative. It is interesting to examine the implications that the pursuit of this goal may have for an organisation. The flexibility of an IT infrastructure depends largely on the extent to which its resources are “sharable and reusable” (i.e. the connectivity of the IT platform and the organisation’s capacity to share different types of IT services). Therefore, a key issue for an organisation is the configuration of its IT platforms, network and telecommunications, and this in turn raises questions concerning configuration, compatibility and

integration rules, access standards, connectivity of systems and excess capacity over the current requirements (Duncan, 1995, p. 42). These needs have led to a move towards distributed computing and standardisation (or “open systems”) that give a high level of connectivity.

One example of a company that has implemented such an infrastructure for purposes of improving BPR success is Sweden Post (Moreton and Chester, 1997). Sweden Post has an IT strategy that not only supports the business operations but also allows a high degree of freedom in the choice of IT application areas. IT is integrated into the majority of its business operations and the IT function provides both cost reduction and competitive edge applications across the organisation. Business managers decide what data is needed to support their requirements at the level of the strategic business unit (SBU). The IT strategy ensures that data can be made available in a standard form (for sharing between the SBU and regions) and with a standard data structure (for collection, processing and presentation). The IT infrastructure that supports these applications ensures the standardisation of computers and communications technology to optimise investment costs and to ensure that data is shared. Sweden Post’s IT strategy also aims to provide for a high degree of independence and flexibility so that the organisational changes associated with this level of transformation can be integrated into the existing processes and structure over time (Moreton and Chester, 1997, pp. 147-9). In short, there are two main implications for organisations that are attempting radical or evolutionary BPR exercises where the role of IT is “proactive”. First, a flexible infrastructure is mandatory, and second there must be dynamic strategic alignment (Avison *et al.*, 2004) between the IT and business strategies of which the BPR programme is a part.

9. Conclusions

The most important contribution of this paper is to demonstrate the ways in which the role of IT can affect the outcome of the BPR effort in a typical organisation. The development of the framework supports this objective. It shows that the different roles of IT provide effective support for different types of BPR, and indicates that aiming for a type of BPR that is not compatible with the present role of IT infrastructure will reduce the probability of success for the BPR project. If this is ignored the BPR effort will either fail or will not produce the level of results that are often expected from BPR projects. At the very least the use of the framework would force the BPR team to address “the IT question” from the very beginning of a programme and to give consideration to the realistic role of IT in the redesign of the business processes. The framework gives some guidelines to the direction in which the IT strategy needs to move if a company is to build an IT infrastructure that is capable of supporting the intended type of BPR project. It is also intended to raise organisational awareness of the issue of IT flexibility and to indicate the importance of taking a long-term view to building IT infrastructure.

The next step for the proposed framework as it exists would be to validate its use in real world situations. A useful area of useful research could then be into the area of BPR “failure”, and case studies of companies that had failed in their attempts at BPR could be analysed to see if these failures could have been predicted from using the proposed framework. Future research in the area would be to develop more detailed guidelines for positioning the present IT infrastructure on the framework and for

determining the flexibility of IT infrastructure. To this end, it may be possible to combine the proposed framework with Duncan's (1995) framework for evaluating the flexibility of the infrastructure. Also, the framework would benefit from the inclusion of practical guidelines on how a firm can move from one role of IT to another, and on which type of BPR is most suited to a particular business change programme. This research would hopefully lead to a better understanding of what is required for implementing successful programmes of business change in terms of the role that IT can play in BPR programmes.

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